Adaptive Score Normalization for Output Integration in Multiclassifier Systems

Giuseppe Pirlo, *Member, IEEE*, and Donato Impedovo, *Member, IEEE*

***Abstract*—This letter introduces a new score normalization technique – based on Dynamic Time Warping (DTW) – for output integration in multiclassifier systems. More precisely, DTW is used to match the score cumulative distribution of each individual classifier against a standard cumulative distribution. The warping function allows optimal alignment of the scores provided by the individual classifiers with the scores on the standard cumulative distribution. Furthermore, in order to adapt the normalization process to the behavior of the individual classifiers and to the decision fusion rule, a new class of fuzzy cumulative distributions is introduced and a genetic approach is used to select the optimal distribution to be used as standard cumulative distribution for score normalization. The experimental tests report better results for the fuzzy normalization technique than for those obtained with other approaches present in the literature.**

***Index Terms*—DTW, fuzzy cumulative distribution, genetic algorithm, multiclassifier system, score normalization.**

I. Introduction

PERFORMANCE of measurement-level multiclassifier systems strongly depends on the effectiveness of the strategy used for normalizing the confidence scores provided by the individual classifiers [1], [2].

Score normalization techniques can be classified into two main categories. The first includes techniques that use no a priori knowledge on the score distribution of the individual classifier. The Min-max technique [3] is an example of normalization technique in this category since the scores provided by the individual classifier are simply scaled linearly. The second exploits some kind of a priori knowledge on score distribution of each classifier. Examples are Z-score [4], Median and MAD [5], Tanh estimators [6], and Characteristic Function [7]. These techniques normalize the scores provided by an individual classifier using statistical estimates of the score distribution of the classifier, as a priori knowledge. Therefore, the result of the normalization process strongly depends on the extent to which the statistical estimates are able to adequately model the score distribution of each individual classifier.

This letter presents a fuzzy technique for score normalization. The technique uses Dynamic Time Warping (*DTW*) to match the score cumulative distribution of each individual classifier against a standard cumulative distribution. The normalized version of a score provided by each individual classifier is then computed according to the couplings identified by the optimal warping functions between the score cumulative distribution of the individual classifier and the standard cumulative distribution. Moreover, in order to select the standard cumulative distribution best suited for the behavior of the set of classifiers and for the decision fusion rule of the combined classifier, a new class of *Fuzzy Cumulative Distributions* (*FCD*) is introduced. A genetic approach is used to define the optimal *FCD* which is used as standard cumulative distribution.

The organization of the letter is as follows. Section II presents a brief survey of score normalization techniques. The new score normalization technique is presented in Section III. Section IV introduces the class of *FCD* and also presents the genetic algorithm for the selection of the optimal *FCD*. In Section V, the experimental results are illustrated. The conclusions of the letter are presented in Section VI.

II. Score Normalization Techniques

In a measurement-level combined classifier, given an unknown pattern x, each individual classifier $\mathrm{A_k}, \mathrm{k} = 1, 2, \ldots, \mathrm{K}$, produces a set of scores

$$\begin{aligned}\mathrm{A_k(x)} &= (\mathrm{A_k}(\mathrm{x},\omega_1), \mathrm{A_k}(\mathrm{x},\omega_2), \ldots, \mathrm{A_k}(\mathrm{x},\omega_\mathrm{i}), \ldots, \mathrm{A_k}(\mathrm{x},\omega_\mathrm{M}))\\ &= (\mathrm{a_1}, \mathrm{a_2}, \ldots, \mathrm{a_i}, \ldots, \mathrm{a_M})\end{aligned}$$

where each $\mathrm{a_i} = \mathrm{A}(\mathrm{x},\omega_\mathrm{i})$ is a confidence value supporting the hypothesis that x belongs to the class $\omega_{\mathrm{i}.}$, being $\Omega = \{\omega_1, \omega_2, \ldots \omega_\mathrm{i}, \ldots, \omega_\mathrm{M}\}$ the set of the classes. Although in the ideal case $\mathrm{a_i}$ is the a-posteriori probability of ω_i given the input pattern x, in many practical cases $\mathrm{a_i}$ is simply the score obtained as the result of a matching process. Therefore, scores provided by different individual classifiers are generally not homogeneous both in terms of range of variability and statistical distribution. Thus, the problem of score normalization is very significant when scores from different classifiers have to be combined [1], [2].

Given a set of matching scores $\mathrm{a_i}, \mathrm{i} = 1, 2, \ldots, \mathrm{M}$, the set of normalized scores $\mathrm{a'_i}, \mathrm{i} = 1, 2, \ldots, \mathrm{M}$ obtained by some of the most widespread techniques are reported in the following:

- Min-max normalization [3]:

$$a'_\mathrm{i} = \frac{a_\mathrm{i} - S_{\min}}{S_{\max} - S_{\min}} \tag{1}$$

  being $\mathrm{S_{min}}$ and $\mathrm{S_{max}}$ the minimum and the maximum of the scores, respectively.
- Z-score normalization [4]:

$$a'_\mathrm{i} = \frac{a_\mathrm{i} - \mu}{\sigma} \tag{2}$$

  where μ is the arithmetic mean and σ is the standard deviation of the given data set $\mathrm{a_i}, \mathrm{i} = 1, 2, \ldots, \mathrm{M}$.

Manuscript received July 17, 2012; revised September 10, 2012; accepted September 12, 2012. Date of publication October 02, 2012; date of current version October 11, 2012. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Tolga Tasdizen.

The authors are with the Dipartimento di Informatica of the University of Bari, 70125 Bari, Italy (email: giuseppe.pirlo@uniba.it).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/LSP.2012.2221708

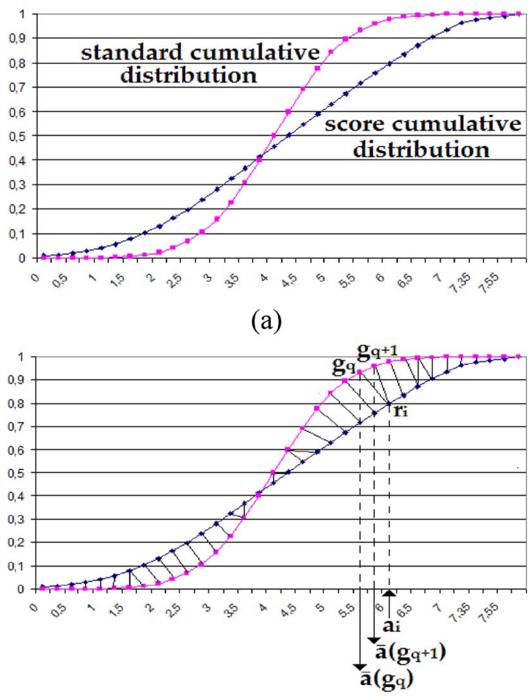

Fig. 1. DTW-based score normalization.

- Median and Median Absolute Deviation (MAD) [5]:

$$a'_i = \frac{(a_i - median)}{(MAD)} \tag{3}$$

where $\mathrm{MAD} = \mathrm{median}\ (\ |a_i - \mathrm{median}|\ )$.

- Tanh estimators [6]:

$$a'_i = \frac{1}{2}\left\{\tanh\left[0.01 \cdot \left(\frac{a_i - \mu}{\sigma}\right)\right] + 1\right\} \tag{4}$$

where μ is the arithmetic mean and σ is the standard deviation of the set of matching scores.

- Characteristic Function [7]:

$$a'_i = a_i + Charf(a_i) \tag{5}$$

where the Characteristic Function of the classifier is defined as:

$$Charf(a_i) = a_{\max} \cdot s_i - a_i \tag{6}$$

being $a_{\max}$ the maximum matching score and s_i the partially accumulated recognition rate, that is defined as:

$$s_i = \frac{\sum\limits_{a_j \le a_i} n_{\mathrm{correct}}(a_j)}{N} \tag{7}$$

with N being the number of patterns and $n_{\mathrm{correct}}(\cdot)$ the correct recognition function of the classifier that counts, for each matching score a_i, the number of correctly recognized samples $n_{\mathrm{correct}}(a_i)$ for which the classifier provides a matching score equal to a_i.

III. Score Normalization Using Dynamic Time Warping

Score normalization can be performed using Dynamic Time Warping (*DTW*) to match the score cumulative distribution of the correct recognitions of an individual classifier against a standard cumulative distribution [8]. More precisely, let

- $\mathrm{R} = (r_1, r_2, \ldots\ldots, r_{\mathrm{M}})$ be the score cumulative distribution of the correct recognitions of an individual classifier,
- $\mathrm{G} = (g_1, g_2, \ldots\ldots, g_{\mathrm{M}})$ be the standard cumulative distribution,

where it is assumed that both distributions were represented by uniformly sampled data sequence and $r_1 = g_1 = 0$ and $r_M = g_M = 1$. Furthermore, let $\mathrm{W(R,G)} = \langle c_1, c_2, \ldots, c_T\rangle$ be a warping function joining elements of R and G, where $c_t = (i_t, j_t)$, being T, i_t, j_t integers, $1 \le t \le T$, $1 \le i_t \le M$, $1 \le j_t \le M$. If a distance $d(c_t) = d(r_{i_t}, g_{j_t})$ is defined between elements of R and G, a dissimilarity measure

$$D_{W(R,G)} = \sum_{t=1}^{T} d(c_t) \tag{8}$$

can be associated to W(R, G). The *DTW* detects the optimal warping function $W^*(R,G) = \langle c_1^*, c_2^*, \ldots, c_{T^*}^*\rangle$ which satisfies the monotonicity, continuity and boundary conditions, and for which it results:

$$D_{W^*(R,G)} = \min_{W(R,G)} D_{W(R,G)}. \tag{9}$$

The normalized value of a score a_i provided by the individual classifier is given by [8]:

$$a'_i = \frac{1}{Card(G_i)} \cdot \sum_{g_q \in G_i} \overline{a}(g_q) \tag{10}$$

where G_i is the set of elements of G joined to r_i by $W^*(R,G)$, (i.e., $G_i = \{g_q | (i,q) \in W^*(R,G)\}$); $\overline{a}(g_q)$ is the score corresponding to the element g_q of G.

Fig. 1(a) shows a schematic example of a score cumulative distribution and of a standard cumulative distribution. The process of DTW-based score normalization is shown in Fig. 1(b). The optimal warping function $W^*(R,G)$ is used to determine the best couplings between r_i, that is the value of the score cumulative distribution which corresponds to the score a_i provided by classifiers, and points of the standard cumulative distribution. In the example r_i is coupled to two points, g_q and g_{q+1}, to which correspond the scores $\overline{a} = (g_q)$ and $\overline{a} = (g_{q+1})$. Then, in accordance with (10), the normalized score a'_i is given by averaging $\overline{a} = (g_q)$ and $\overline{a} = (g_{q+1})$ as follows: $a'_i = (1/2) \cdot [\overline{a} = (g_q) + \overline{a} = (g_{q+1})]$. Since the standard cumulative distribution is the same for all individual classifiers, this technique allows the normalization of the scores of all classifiers on a common scale [8]. It is worth noting that the potential of the DTW-based score normalization technique can be fully exploited by selecting the optimal standard cumulative distribution to be used, depending on the behavior of the individual classifiers and the decision fusion rule of the combined classifier. The next Section presents an original formulation of the optimization problem for selecting the best fuzzy cumulative distribution for score normalization.

IV. Fuzzy Cumulative Distributions for Adaptive Score Normalization

In this Section, a new class of fuzzy distributions, named *Fuzzy Cumulative Distributions (FCD)*, is introduced in order to adapt the DTW-based score normalization technique to the behavior of the individual classifiers and to the decision fusion rule of the combined classifier. An *FCD* is here defined as any sequence $\{\mu_1, \mu_2, \ldots \mu_j, \ldots, \mu_M\}$ of M values (M integer) for which the following properties hold:

- boundary conditions:

$$\mu_1 = 0 \text{ and } \mu_M = 1. \tag{11a}$$

- monotonicity:

$$\mu_m \leq \mu_{m+1},\ m = 1, 2, \ldots, M - 1. \tag{11b}$$

The selection of the best suited FCD for a given set of individual classifiers most certainly involves detecting the optimal values μ_m, $m = 1, 2, \ldots M$, which maximize the classification performance of the combined classifier. In this work we considered the Error Rate (ER) of the combined classifiers as the cost function associated to the optimization problem. The optimization problem for the selection of the best FCD was then defined as follows:

$$\begin{aligned}&\text{Find FCD}^* = \{\mu_1^*, \mu_{2,}^* \ldots \mu_j^*, \ldots, \mu_M^*\} \text{ so that :}\\ &\text{ER}(\text{FCD}^*) = \min_{\{\text{FCD}\}} \text{ER}(\text{FCD}).\end{aligned} \tag{12}$$

What follows is a description of the real-coded genetic technique used for the optimization problem of (12) [9]:

STEP 1. The initial–population $\text{Pop} = \Phi_1, \Phi_2, \ldots, \Phi_l, \ldots, \Phi_{\text{Npop}}$ was created by generating N_{pop} random individuals (N_{pop} even), where each Φ_i was a FCD that was defined through the sequence of the values $\Phi_i = (\mu_{i,1}, \mu_{i,2}, \ldots, \mu_{i,m}, \ldots, \mu_{i,M})$.

STEP 2. The cost function of an individual $\Phi_i = (\mu_{i,1}, \mu_{i,2}, \ldots, \mu_{i,m}, \ldots, \mu_{i,M})$ was taken as the Error Rate of the combined classifier, when Φ_i was considered as the standard cumulative distribution of the score normalization process.

STEP 3. From the current – population, the new populations of individuals were generated by the following genetic operators:

- *Selection.* The roulette-wheel strategy was used to select $N_{pop}/2$ random pairs of individuals for crossover.
- *Crossover.* The one-point crossover was used to combine information from different individuals. Precisely, let

$$\Phi_i = (\mu_{i,1}, \mu_{i,2}, \ldots, \mu_{i,s}, \mu_{i,s+1}, \ldots, \mu_{i,M}) \tag{13a}$$

and

$$\Phi_j = (\mu_{j,1}, \mu_{j,2}, \ldots, \mu_{j,s}, \mu_{j,s+1}, \ldots, \mu_{j,M}) \tag{13b}$$

be two individuals selected for crossover, and let s be a random integer, $1 \leq s < M$, the two offspring individuals Φ_i' and Φ_j' of the next generation were defined as:

$$\Phi'_i = (\mu_{i,1}, \mu_{i,2}, \ldots, \mu_{i,s}, \mu_{j,s+1}, \ldots, \mu_{j,M}) \tag{13c}$$

and

$$\Phi_j' = (\mu_{j,1}, \mu_{j,2}, \ldots, \mu_{j,s}, \mu_{i,s+1}, \ldots, \mu_{i,M}). \tag{13d}$$

- *Mutation.* Given an individual $\Phi_i = (\mu_{i,1}, \mu_{i,2}, \ldots, \mu_{i,m}, \ldots, \mu_{i,M})$ the mutation operator worked as follows. By letting $\mu_{i,m}$ be an element of Φ_i selected for mutation, according to a mutation probability $\mu\_\text{Mut\_Prob}$, the mutation operator changed $\mu_{i,m}$ into $\mu_{i,m}'$, with:

$$\mu_{i,m}' = \mu_{i,m+(-1)}^{e} \cdot \eta,\ m = 1, 2, \ldots,\ M \tag{14}$$

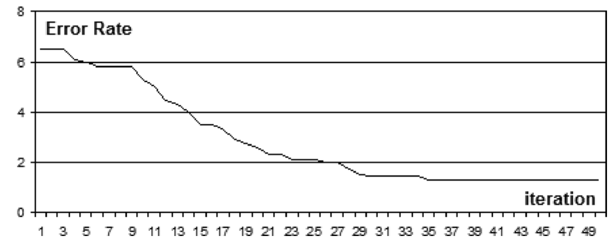

Fig. 2. Optimal fuzzy cumulative distribution.

where e was a random Boolean value generated according to a equally-distributed probability function; η was a random value generated in the range $[0, \eta\_\max]$, according to a uniform distribution.

After the mutation operator had been applied to an individual, the following rules were used to guarantee that the properties (11) were satisfied:

a) boundary conditions:

$$\text{Set } \mu_{i,1}' = 0 \text{ and } \mu_{i,M}' = 1; \tag{15a}$$

b) Monotonicity:

$$\begin{aligned}\text{For } m = 1, 2, \ldots, M \text{ do : if } \mu_{i,m}' > \mu_{i,m+1}' \text{ then}\\ \text{Set } \mu_{i,m+1}' = \mu_{i,m}'.\end{aligned} \tag{15b}$$

- *Elistism.* One random individual of the current population was substituted with the best individual of the previous population.

In the genetic process, steps 2 and 3 were repeated until at least one of the following stop conditions occurred: (a) N^{iter} successive populations were generated; (b) no improvement was registered in T successive populations. Of course, when a stop condition occurred, the process stopped and the optimal FCD to be used for score normalization was obtained from the best individual of the last-generated population.

V. Experimental Results

The experimental tests were conducted in the field of handwritten digit classification. The digit database of the CEDAR was utilized for training, validation and testing [10]. From 18,467 patterns of the BR directory, 14,467 were used for training the individual classifiers and the remaining 4,000 for selecting the score normalization parameters. For the test, 2,189 patterns of the BS directory were divided into four test folders: TF1, TF2, TF3, TF4 (the number of patterns in each folder was 550, 550, 550 and 539, respectively). The following parameters were used for the genetic algorithm adopted for selecting the optimal FCD [10]: $M = 50$; $N_{Pop} = 100$; $N^{iter} = 50$; $T = 5$; $\mu\_\text{mut\_prob} = 0.30$; $\eta\_\max = 0.02$. The Euclidean distance was considered for the DTW procedure. The multiclassifier system used the sum-rule to combine three k-nn ($k = 1$) individual classifiers [8]: A_1 used 14 geometrical features, A_2 used 32 contour features, and classifier A_3 used 20 intersection features. Fig. 2 shows the convergence of the genetic algorithm as the number of iteration increases.

Fig. 3 shows the score cumulative distributions of A_1, A_2 and A_3, and the optimal fuzzy cumulative distribution selected by the genetic algorithm.

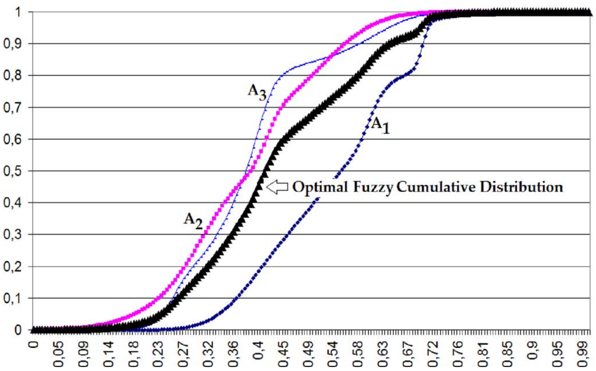

Fig. 3. Optimal fuzzy cumulative distribution.

TABLE I
INDIVIDUAL CLASSIFIERS: RECOGNITION RATE

Individual Classifier	TF1	TF2	TF3	TF4
A_1	93.42%	94.22%	93.57%	94.00%
A_2	89.84%	90.00%	88.75%	89.48%
A_3	95.69%	96.12%	95.61%	96.20%

Table I reports the recognition rate of the individual classifiers on the test patterns. On average, the recognition rate of A_1, A_2 and A_3 is equal to 93.80%, 89.52% and 95.91%, respectively.

Table II reports the recognition rate of the combined classifier on the test patterns. Regarding the standard techniques, it was observed that the Z-score technique provided the best results on average. In fact, when Z-score was used, the average recognition rate was 98.06%. The Min-Max, DTW-based and Characteristic Function techniques, on average led, respectively, to recognition rates of 97.95%, 97.91%, 97.88%. Instead, the Tanh estimators and the Median and MAD techniques led to an average recognition rate equal to 96.24% and 95.16%, respectively. When no technique was considered for score normalization a recognition rate of 95.72% was obtained on average. It is quite interesting to observe that, when no score normalization technique was used, the recognition rate of A_3 was better then the recognition rate provided by the combined classifier. On the contrary, several score normalization techniques provided very good results. The Characteristic Function technique was the best when TF1 and TF3 were considered (the recognition rate was equal to 97.95% and 97.88%, respectively), Z-Score performed best when TF2 was used (giving a recognition rate equal to 98.53%), DTW-based normalization provided the best result when TF4 was used (showing a recognition rate equal to 98.24%). Table II also reports the results provided by the new technique based on DTW along with the optimal fuzzy cumulative distribution. The results show the new technique outperforms the other standard techniques for score normalization in each of the test folders. On average, the new technique led to a recognition rate of 98.69% in all four test folders.

VI. CONCLUSION

This letter presents a new technique for fuzzy score normalization and demonstrates its superior performance with respect to standard techniques for score normalization. The technique uses DTW to match the score cumulative distribution of correct recognitions of each individual classifier against a standard cumulative function – that is the same for all individual classifiers – in order to suitably normalized scores to a common domain. In addition, a new class of *Fuzzy Cumulative Distributions* is defined and a genetic approach is proposed to select the optimal fuzzy distribution to be used for score normalization.

The experimental result shows the advantage of the new technique with respect to the performance of standard approaches presented in the literature.

TABLE II
INDIVIDUAL CLASSIFIERS: RECOGNITION RATE

Score Normalization	TF1	TF2	TF3	TF4
NONE	95.32%	96.42%	95.54%	95.61%
Min-Max [3]	97.80%	98.39%	97.51%	98.10%
Z – Score [4]	97.80%	98.53%	97.73%	98.17%
Median and MAD [5]	95.17%	95.69%	94.37%	95.39%
Tanh Estimators [6]	96.05%	96.49%	95.98%	96.42%
Char. Function [7]	97.95%	98.39%	97.88%	97.29%
DTW (Gaussian)[8]	97.37%	98.46%	97.58%	98.24%
DTW (FCD)	98.34%	99.16%	98.71%	98.55%

REFERENCES

[1] A. Jain, K. Nandakumar, and A. Ross, "Score normalization in multimodal biometric systems," *Patt. Recognit.*, vol. 38, pp. 2270–2285, 2005.
[2] L. Xu, A. Krzyzak, and C. Y. Suen, "Methods of combining multiple classifiers and their applications to handwriting recognition," *IEEE Trans. Syst., Man, Cybern.*, vol. 22, no. 3, pp. 418–435, 1992.
[3] S. Tulyakov, S. Jaeger, V. Govindaraju, and D. Doermann, "Review of classifier combination methods," in *Machine Learning in Document Analysis and Recognition – Studies in Computational Intelligence*. Berlin, Germany: Springer, 2008, vol. 90, pp. 361–386.
[4] H. J. Jung and M. Lease, "Improving consensus accuracy via Z-score and weighted voting," in *Human Computation, Proc. 3rd Human Computation Workshop (HCOMP) at AAAI*, San Francisco, CA, Aug. 7–11, 2011, pp. 88–90.
[5] J. Kittler, M. Hatef, R. P. W. Duin, and J. Matias, "On combining classifiers," *IEEE Trans. Patt Anal. Mach. Intell.*, vol. 20, no. 3, pp. 226–239, 1998.
[6] F. R. Hampel, P. J. Rousseeuw, E. M. Ronchetti, and W. A. Stahel, *Robust Statistics: The Approach Based on Influence Functions*. New York: Wiley, 1986.
[7] O. Velek, S. Jaeger, and M. Nakagawa, "A new warping technique for normalizing likelihood of multiple classifiers and effectiveness in combined on-line/off-line Japanese character recognition," in *Proc. 8th IWFHR*, Niagara-on-the-Lake, ON, Canada, 2002, pp. 177–182.
[8] R. Modugno, G. Pirlo, and D. Impedovo, "Score normalization by dynamic time warping," in *Proc. CIMSA*, Italy, Sep. 6–8, 2010, pp. 82–85.
[9] T. Baeck, *Evolutionary Algorithms in Theory and Practice: Evolution Strategies, Evolution Programming, Genetic Algorithms*. New York: Oxford Univ. Press, 1996.
[10] J. Hull, "A database for handwritten text recognition research," *IEEE Trans. Patt Anal. Mach. Intell.*, vol. 16, no. 5, pp. 550–554, 1994.